

Using machine learning and clinical registry data to uncover variation in clinical decision making

Charlotte James^{a,*}, Michael Allen^{a,b}, Martin James^{a,c}, Richard Everson^d

^a College of Medicine and Health, University of Exeter, Exeter, UK

^b NIHR South West Peninsula Applied Research Collaboration (ARC), UK

^c Royal Devon and Exeter NHS Foundation Trust, Exeter, UK

^d Department of Computer Science, University of Exeter, Exeter, UK

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ABSTRACT

Clinical registry data contains a wealth of information on patients, clinical practice, outcomes and interventions. Machine learning algorithms are able to learn complex patterns from data. We present methods for using machine learning with clinical registry data for quality improvement by identifying where variation in decision making occurs. Using a registry of stroke patients, we demonstrate how machine learning can be used to: investigate whether patients would have been treated differently had they attended a different hospital; group hospitals according to clinical decision making practice; identify where there is variation in decision making between hospitals; characterise patients that hospitals find it hard to agree on how to treat. Our methods should be applicable to any clinical registry and any machine learning algorithm to investigate the extent to which clinical practice is standardized and identify areas for improvement at a hospital level.

1. Introduction

Clinical registries are collections of information on people that have a specific disease, have experienced, or are at-risk of, an adverse health-related event or have been exposed to something that might have an adverse effect. They provide information on clinical practice, patient outcomes and the effect of interventions and are therefore useful for monitoring the course of disease, examining the effect of a treatment on patient outcome and measuring quality of care [1].

In England the Healthcare Quality Improvement Partnership (HQIP), on behalf of the National Health Service (NHS), is responsible for overseeing and commissioning more than 30 clinical audits (registries), which form the National Clinical Audit Programme. These collect and analyse registry data supplied by clinicians and are intended to be used for quality improvement by allowing the quality of care and services to be measured against agreed standards and for improvements to be made where necessary [2]. The quantity and quality of data contained in these registries has the potential to provide insight into additional processes that can have an impact on patient outcomes, such as clinical decision making. As decision making processes are complex and involve many interacting factors, standard statistical analyses are unlikely to capture this insight.

Machine learning algorithms are able to learn complex patterns from data. They have shown promise in many areas, including drug discovery [3], precision medicine [4], medical diagnosis [5,6], and clinical decision support [7,8] suggesting that, given the right data, ML algorithms can learn clinical decision making processes. As the data contained in clinical registries includes information on patients, clinical practice, treatment and outcomes, ML presents an opportunity to understand the decision making processes of clinicians, assess the extent to which these processes are standardized and, if not, uncover where decision making varies.

The national audit covering stroke is the Sentinel Stroke National Audit Programme (SSNAP) [9e]. SSNAP collects longitudinal data on the processes and outcomes of stroke care up to 6 months post-stroke for more than 95% of emergency stroke admissions to acute hospitals in England, Wales and Northern Ireland. Every year data from approximately 85,000 patients are collected. SSNAP publishes quarterly and yearly analyses of results on its website. Data from SSNAP has been previously used to investigate clinical decision making [10], socioeconomic risk factors for stroke and their influence on care received and patient outcomes [11], and temporal variation in stroke care [12].

Stroke is a leading cause of death and disability worldwide which occurs when there is a blood clot or bleed in the brain preventing blood

* Corresponding author.

E-mail address: charlotte.james@bristol.ac.uk (C. James).

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from reaching parts of the brain. Around 80–85% of strokes are due to a blood clot (ischemic stroke). Thrombolysis is the only licensed drug treatment for acute ischemic stroke. It is critically time dependent: the longer the time window between stroke and administration, the less benefit to the patient. After 4.5 h there is little or no benefit of thrombolysis [13]. In England, Wales and Northern Ireland, 11.1% of patients with confirmed acute stroke receive thrombolysis, however there is considerable variation between hospitals, from 0% to 24.5%. Reasons for this variation are known to include slow uptake of the treatment and in-hospital delays to administration [14]; it is also possible that differences in clinical decision making play a part [15].

To investigate whether variation in thrombolysis rates between hospitals is due, at least in part, to differences in clinical decision making we used a ML approach. Specifically, we trained ML algorithms to learn the decision making outcome in each hospital. The algorithms are trained using data that is available to the clinician and that the clinician would use to decide whether or not to administer thrombolysis to a patient. The algorithms predict the binary outcome, to thrombolysed (1) or not (0). By using ML to learn the decision making processes in a hospital, we are able to answer counterfactual, or ‘what if?’, questions, such as ‘what if a patient had been admitted to a different hospital?’. This allows us to determine whether differences in clinical decision making contribute to the variation in thrombolysis rates observed between hospitals, and characterise patients that hospitals find it harder to agree how to treat.

While we focus on clinical decision making surrounding the use of thrombolysis for acute stroke, the methods we present should be applicable to any clinical registry data. These methods could be used with clinical registries to retrospectively assess variation in decision making between hospitals, group hospitals according to their decision making and characterise patients that clinicians find it difficult to agree how to treat. Uncovering differences in clinical practice using these methods may provide an opportunity to improve patient care by standardizing treatment across the board.

2. Materials & methods

2.1. Data

We used data from SSNAP collected over 3 years, from January 2016 to December 2018, corresponding to 246,676 emergency stroke admissions to 180 acute care hospitals. We restricted our analysis to hospitals that admitted at least 100 patients during the 3 year time period and thrombolysed 10 or more. As thrombolysis is only considered to have a significant effect if given within 4.5 h of stroke onset [13] and we are interested in emergency stroke admissions, only patients that had a stroke out of hospital and arrived in hospital within 4 h of known stroke onset were included in the analysis. This resulted in 132 unique hospitals and 88,928 stroke patients, of which 26,257 (29.5%) received thrombolysis.

The data contained 47 features: admitting hospital ($N = 1$), patient characteristics ($N = 5$), pathway information ($N = 10$), comorbidities ($N = 10$), National Institutes of Health Stroke Scale (NIHSS) ($N = 16$), other clinical information ($N = 4$) and whether thrombolysis was given ($N = 1$) (Table S1). The NIHSS is a 15 item neurological scale that is used by clinicians as a measure of stroke severity. It ranges from 0 to 42 and may be used to classify a stroke as mild (NIHSS 1–4), moderate (NIHSS 5–14), severe (NIHSS 15–24) or very severe (NIHSS >25) [16]. Also present in the data is a patient’s modified Rankin Scale (mRS) score before stroke, which is a measure of pre-stroke disability ranging from 0 to 5, with 0 corresponding to no prior disability and 5 meaning severely disabled.

Of the 47 features, 24 were missing for at least one patient (Table S2). If components of the NIHSS were missing they were replaced with 0 and if the time from arrival to brain scan (*Brain Imaging Time (minutes)*) was missing it was replaced with 9999, both of which force the probability of thrombolysis to be low. If a missing feature was binary

it was coded as ‘missing’. All binary and categorical variables were one-hot-encoded, with ‘missing’ treated as a separate category.

2.2. Learning clinical decision making

We used a ML approach to learn clinical decision making outcomes from the data.

Specifically, we implemented a Random Forest (RF) which is an example of an ensemble learning algorithm composed of individual decision trees. We used RF as it is known to be robust and have a lower variance compared to a single decision tree. During training a RF will select a random sample of the training data with replacement and fit a decision tree to the sample. This process is repeated many times, the exact number being a parameter of the algorithm, to create an ensemble (or forest) of decision trees [17].

The RF is trained to perform a binary classification task: to decide whether a patient should be thrombolysed (class 1) or not (class 0). Each decision tree in the RF performs this classification task independently. A patient’s final class is determined by averaging over the outcomes of all decision trees: if the average is less than the decision threshold, the patient is assigned class 0 (not thrombolysed) else the patient is assigned class 1 (thrombolysed).

To implement the RF, we used Python’s sci-kit learn library [18]. We set aside a representative cohort of 10,000 patients for testing and used the remaining data, representing 78,928 patients, for model development. Using the development data, we implemented 10-fold cross-validation: the data was split into a training set (90%) and validation set (10%) ten times, such that each patient is in the validation set exactly once. The RF was fit to the training set at each fold and the validation set was used to assess performance. Performance measures were determined by averaging over those for each fold. The RF was subsequently fit to all 78,928 patients, without cross-validation, and performance was assessed by applying the models to the representative cohort.

2.3. Hospital models

We divided our data into separate hospitals and trained an RF on patients from each hospital. Each of these RFs learn the decision making within a single hospital during training. We again used stratified 10-fold cross-validation for model development and assessed how well each model was able to learn the decision making at each hospital. The decision threshold of each model was calibrated such that the predicted positive rate (i.e. fraction of patients predicted to be given thrombolysis) was equal to the true positive rate. Final performance was assessed by fitting each hospital model to the development set without cross-validation and applying each model to patients from the same hospital in the test data.

2.4. Assessing variation in decision making

To investigate how decision making varies between hospitals, we trained RFs on patients from each hospital, combining the test and development data and using all patients from that hospital for training. We subsequently took all other patients in our data set and applied each hospital’s RF to every patient: the classification given by each RF represents what would have happened if a patient had been treated at that hospital.

2.5. Grouping hospitals by decision making

To find hospitals with similar decision making processes, we used the hospital models trained on the development set only. Each RF was applied to all patients in the test set to determine whether or not each patient would have been thrombolysed had they attended each hospital. We computed the pairwise Hamming distance, D_H , between the

predicted outcomes for every pair of hospitals. The Hamming distance is the proportion of patients on which the two hospital models disagree on the outcome: if D_H is close to 0 the hospitals would make similar decisions, whereas if it is close to 1 their modelled decision making processes are different.

3. Results

3.1. Population characteristics

Population characteristics are displayed in Table 1. Patients that were thrombolysed were, on average, younger (mean 73 years), arrived at the hospital in less time (mean 97 min), had brain imaging scans faster (mean 23 min), had more severe stroke (mean NIHSS 11.6) and lower levels of disability (mean rankin 0.7) compared to those that were not thrombolysed.

3.2. Learning clinical decision making

Model performance is described in Table 2: the RF trained on all patients in the development set performed well with a mean accuracy of 84.3% (range 84.0–84.7) and AUC of 0.91 (range 0.91–0.92) across 10 folds (Development) and an accuracy of 83.8% and AUC of 0.91 in the test set. An accuracy of 85% means that 15% of patients were misclassified (Combined Model, Table 2). If clinical decision making was consistent both within and between hospitals we would expect misclassified patients to lie close to the decision boundary: these patients represent borderline decisions. To verify this, we looked at the probability distribution for misclassified patients, where the probability is the average outcome over all trees in the RF. From Fig. 1 it is clear that while the distribution is approximately centred on the decision threshold there is a significant proportion of patients who were confidently misclassified: the probability of receiving thrombolysis is close to 0 or 1. This shows that not all misclassified patients represent borderline decisions and therefore supports our hypothesis that there is variation in clinical decision making: a confident misclassification means that there was a similar patient in the training data who was treated differently.

3.3. Hospital models

In order to understand sources of variation in decision making between hospitals, we trained separate RFs on the patients from each of the 132 hospitals. On average, individual hospital models perform similarly to the combined model in the development data (Table 2). However, for each measure there is a range of values: the lowest performing hospital model has an accuracy of 76.1% and AUC of 0.80, which is significantly lower than the accuracy and AUC of the combined model (84.3% and

Table 1

Population characteristics stratified by outcome for patients in the treatment pathway.

Measure ^a	Thrombolysed N = 26257	Not Thrombolysed N = 62671
Age (years)	73 (14)	76 (14)
Female (N, %)	12084 (46)	30796 (49)
White (N, %)	22861 (87)	55660 (89)
Onset to Arrival (minutes)	97 (46)	117 (54)
Arrival to Imaging (minutes)	23 (20)	157 (1095) ^b
NIHSS	11.6 (7.0)	8.4 (8.4)
Facial Palsy	1.2 (0.9)	0.8 (0.8)
Dysarthria	1.0 (0.8)	0.7 (0.8)
Best Language	1.1 (1.1)	0.7 (1.1)
Rankin before Stroke	0.7 (1.2)	1.2 (1.5)

^a Values correspond to mean (std) unless otherwise stated.

^b Standard deviation affected by imputation method.

Table 2

Performance measures of Random Forest algorithm trained on data from all hospitals together with one-hot encoding of hospital ID (Combined Model) and each hospital separately (Hospital Models).

Performance Measure	Combined Model		Hospital Models	
	Development ^{a,b}	Test	Development ^c	Test ^c
Accuracy	84.3 (84.0–84.7)	83.8	84.2 (73.5–93.0)	81.6 (69.0–87.3)
Sensitivity	0.69 (0.67–0.70)	0.68	0.72 (0.45–0.88)	0.52 (0.15–0.90)
Specificity	0.91 (0.90–0.91)	0.90	0.88 (0.77–0.96)	0.93 (0.73–1.00)
False positive rate	0.09 (0.09–0.10)	0.10	0.12 (0.04–0.23)	0.07 (0.00–0.27)
False negative rate	0.31 (0.30–0.33)	0.32	0.28 (0.12–0.55)	0.48 (0.10–0.85)
Positive predictive value	0.69 (0.67–0.70)	0.68	0.72 (0.45–0.88)	0.52 (0.15–0.90)
Negative predictive value	0.91 (0.90–0.91)	0.90	0.88 (0.77–0.99)	0.93 (0.73–1.00)
Positive rate	0.27 (0.26–0.27)	0.27	0.29 (0.07–0.52)	0.19 (0.06–0.37)
AUC ^d	0.91 (0.91–0.92)	0.91	0.90 (0.78–0.97)	0.84 (0.64–0.91)

^a For a decision threshold of 0.5.

^b Values correspond to mean (range) of 10 folds.

^c Values correspond to mean (range) of all 132 hospital models.

^d AUC independent of decision threshold.

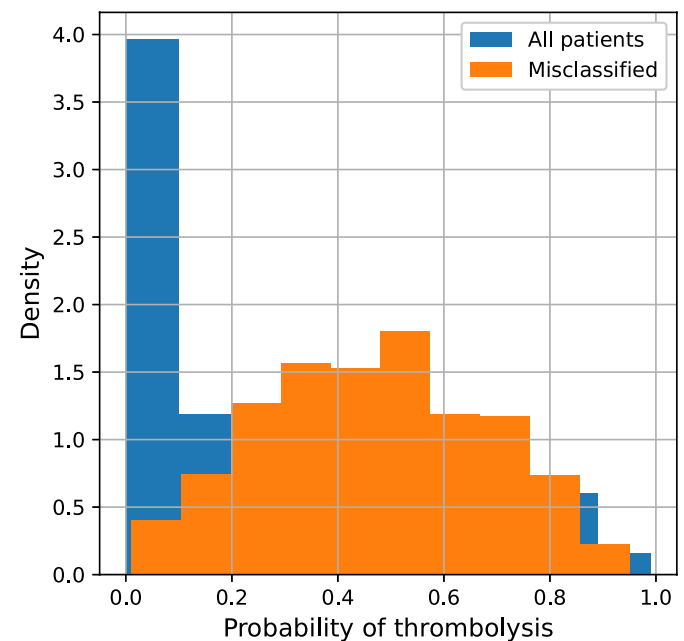


Fig. 1. Probability distribution for patients misclassified by a Random Forest algorithm trained on data from all hospitals together.

0.91). The range of values of the performance measures may be explained by variation in hospital size and thrombolysis rates. The performance of each hospital model will be affected by both the amount of data (hospital size) and class balance (thrombolysis rate): hospitals with a smaller number of patients and lower thrombolysis rate are less likely to have well calibrated models.

We assessed how well hospital models were calibrated by plotting a calibration curve for each model (Fig. S1a). If a model is perfectly calibrated, the proportion of a set of patients that are assigned an equal probability of receiving thrombolysis by the model, that go on to receive thrombolysis, should be equal to the probability: i.e. out of all patients the model assigns a 60% chance of receiving thrombolysis to, 60% of the

patients should receive it.

From Fig. S1a, most hospital models are well calibrated, however some do not output probabilities over the whole range, or are not well calibrated at higher probabilities (Fig. S1b). In order to improve model calibration we applied Platt Scaling [19], fitting a logistic regression to the probabilities output by each hospital model. After applying Platt Scaling, the model calibration curves are closer to that of the perfectly calibrated model (black dashed line, Fig. S1c), however some hospital models remain not well calibrated. Fig. S2 shows how hospitals with models that were not well calibrated compared to other hospitals in terms of number of patients and thrombolysis rate. From this it is clear that hospitals that have both fewer patients and lower thrombolysis rates are less likely to be well calibrated. This may be attributed to the training data for these hospital models being both smaller in size and more imbalanced.

After calibrating the hospital models we assessed their performance in the test set (Table 2, Test). On average, there was a decrease in performance compared to both the combined model and the hospital models in the development set: in the test set hospital models had a mean accuracy of 81.6% and mean AUC of 0.84, with the lowest performing hospital having an accuracy and AUC of 76.1% and 0.64 respectively. From Fig. S3 this decrease in performance can again be attributed to the amount of data available during training (hospital size) and class balance (thrombolysis rate).

As with the combined model, hospital models also confidently misclassify patients in the test set (Fig. S4). In this case, confident misclassifications represent cases where a patient was treated differently to what would be expected in that hospital: for a confident misclassification to occur, a similar patient that was treated differently must be present in the training data. However, the reason that these patients were treated differently to the norm for the hospital they attended is likely not captured in the data and will not be explored further.

3.4. Assessing variation in decision making

Using hospital models, each trained on all patients from a single hospital, we determined how each patient in the dataset would have been treated had they attended a different hospital. We found that it is

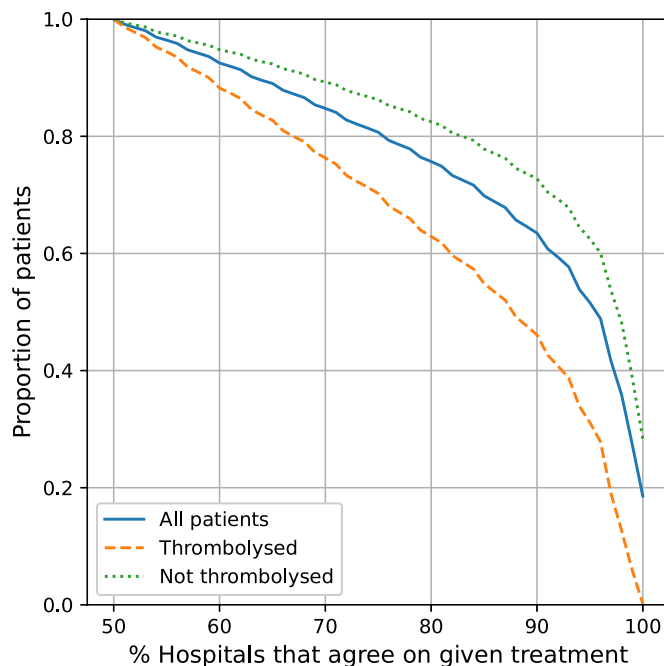


Fig. 2. Proportion of patients that hospitals agree to treat, stratified by patient outcome.

much easier for hospitals to agree who not to thrombolysed than who to give thrombolysis to (Fig. 2). Specifically 80% of hospitals agree not to thrombolysed 85% of patients that did not receive thrombolysis, whereas only 60% of patients who did receive thrombolysis would have received it at 80% of hospitals. Only 3% of patients that were given thrombolysis would have received it at every hospital.

3.5. Grouping hospitals by decision making

To understand where the variation in decision making occurs, and to group hospitals according to their decision making, we used the hospital models trained on the development set and the decisions made in the test set. Computing D_H between the predicted outcomes of each pair of hospitals we found no clear groups of hospitals (Fig. S5), due to the fact that hospitals find it easy to agree on who not to thrombolysed.

Accounting for this, we re-calculated D_H for each pair of hospitals using only contentious patients: patients that would have been thrombolysed at 30–70% of hospitals, of which there were 1,426, reasoning that it is these borderline cases that hospitals with similar decision making processes would make similar decisions on. We found two distinct groups of hospitals (Fig. 3). Looking at the thrombolysis rate in contentious patients for each of these groups, the largest group (location 75–95 in Fig. 3) has a high thrombolysis rate in this group of patients, with a mean of 90%, suggesting that these hospitals are more likely to make the decision to thrombolysed when others may consider it risky. In contrast, the smaller group in the top left of the figure has a low thrombolysis rate, treating on average 16% of contentious patients: these are hospitals that are less likely to take risks when the decision to thrombolysed is borderline.

3.6. Characterising patients where decision making differs

We characterized the group of contentious patients by comparing them to patients that were not contentious: patients that over 70% of hospitals agreed on the treatment of. We found that contentious patients did not differ significantly in terms of age, arrival time, scan time or stroke severity (NIHSS) compared to patients that over 70% of hospitals would have thrombolysed (Table 3). However there was a difference in terms of disability: the modified rankin score of contentious patients was 1.1 ± 1.4 where as for those patients that hospitals agreed to thrombolysed, this was 0.6 ± 1.1 , and for patients hospitals agreed not to

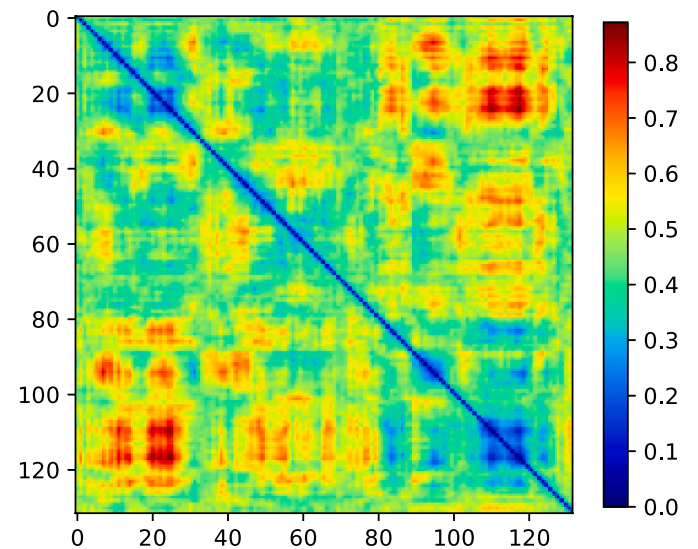


Fig. 3. Pairwise Hamming distance, D_H , between hospitals, derived from the decisions made at each hospital on how to treat a standardized cohort of 10,000 patients.

Table 3

Key measures for patients in a cohort of 10,000 that over 70% of hospitals would have thrombolysed (Thrombolysed), over 70% of hospitals wouldn't have thrombolysed (Not Thrombolysed) and those that were contentious (30–70% of hospitals would have thrombolysed).

Measure ^a	Thrombolysed	Not Thrombolysed	Contentious
Age	73 (14)	76 (14)	75 (14)
Onset to Arrival	89 (41)	123 (55)	101 (47)
Arrival to Imaging	19 (12)	150 (890)	26 (24)
NIHSS	13 (7)	6 (7)	12 (8)
Rankin before Stroke	0.6 (1.1)	1.2 (1.5)	1.1 (1.4)

^a Values correspond to mean (std).

thrombolysed it was 1.2 ± 1.5 . This suggests patients who have a prior disability represent borderline decisions in terms of whether or not thrombolysis should be given, and more severe disability reduces the chance of thrombolysis.

To further understand the difference between contentious patients and those that over 70% of hospitals would have thrombolysed, we trained an additional RF to classify patients into these groups. The RF again performs a binary classification task to determine whether a patient would have been thrombolysed (class 0) or was contentious (class 1). The performance of the RF is a measure of how easy it is to distinguish between these two groups of patients, and determining which features are important for the classification provides insight into how these groups of patients differ.

The RF was able to classify patients with an accuracy of 88% (decision threshold of 0.5) and an AUC of 0.96, suggesting that contentious patients are distinguishable from those that 70% of hospitals would have thrombolysed. To understand how these patients differ, we performed a principal component analysis by re-training the RF with an increasing number of features, starting with the most important. We found that the AUC of the classifier reaches 0.96 with less than 20 features (Fig. 4). Of these features, the most important are NIHSS on arrival and whether the stroke onset time is known or an estimate. From Fig. 5a it is clear that patients that are contentious are more likely to have an estimated onset time than those that hospitals agree to thrombolysed. Fig. 5b shows, for each NIHSS score, the proportion of patients in each group. Here it is clear that hospitals agree not to thrombolysed patients with very mild stroke, where as contentious patients have moderate to severe stroke (NIHSS 5–30). This pattern is also clear from Fig. 5c, where we compare the proportion of patients that hospitals agree how to treat (thrombolysed or not thrombolysed), to the contentious group for each NIHSS score. The number of patients with NIHSS > 30 is very low (Fig. 5d) therefore the distributions in Fig. 5b and c at higher NIHSS values are less representative of the true distribution, as shown by the larger margin of uncertainty in Fig. 5c.

For the RF, feature importance is determined by the Gini importance: for each tree a feature's importance is the total decrease in impurity that occurs when the feature is used to split a node, weighted by the number of samples the node splits. The Gini importance is calculated for each tree and then averaged to give a final feature importance. Shapley values are an alternative method for determining how each feature contributes to a classification. While Gini importance gives an overall measure of importance for each feature, Shapley values contain information on how the value of a feature contributes to the classifier's final decision. Computing Shapley values themselves is computationally intense, however SHAP (SHapley Additive exPlanations) [20] values provide an estimation of Shapley values and can be used to explain how features contribute to model predictions.

Fig. 6 displays the SHAP values for the 20 features that have the highest impact on the output of the RF trained to distinguish between patients that hospitals agree to thrombolysed and those that are contentious. For each feature on the y-axis, each point represents the SHAP value of that feature for a single patient. The SHAP value itself (x-axis) tells us whether the feature pushed the model towards classifying a

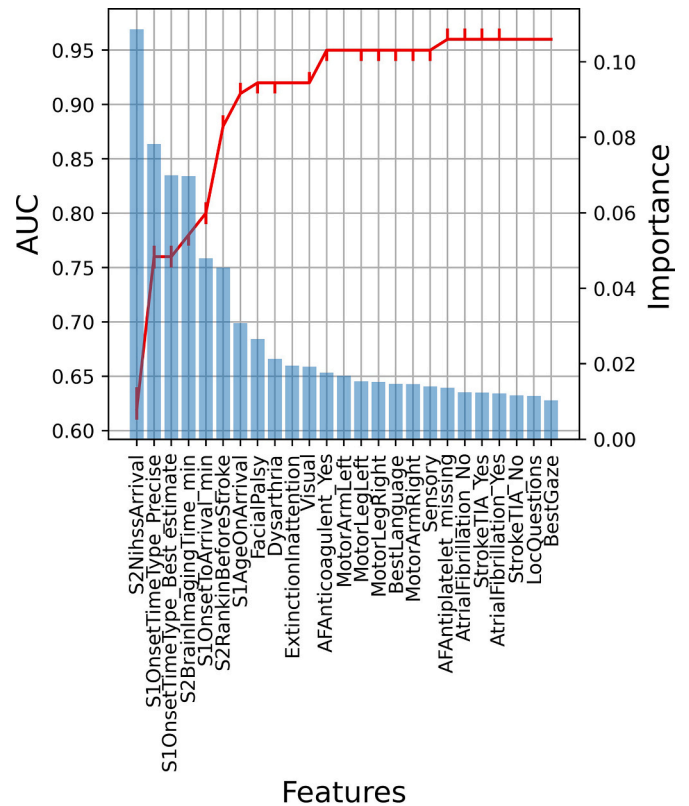


Fig. 4. Features, ordered by importance (x-axis) vs importance (y-axis right) and cumulative AUC (y-axis left) for a random forest trained to classify whether hospitals would agree to thrombolysed a patient or whether they are contentious. Red line represents the median AUC and error bars correspond to 10th and 90th percentiles after bootstrapping the training data 100 times. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

patient as contentious (class 1, positive SHAP) or not (class 0, negative SHAP). For example, from Fig. 6 the most important feature in determining whether a patient would be thrombolysed, or is contentious, is whether or not the onset time is precise. This is a binary feature, where a low value (ie, 0) has a positive SHAP value, corresponding to the onset time being uncertain and pushing a patient towards the contentious class. In contrast, a high value (ie, 1) represents onset time being precisely known and the corresponding SHAP value is negative, pushing a patient towards the class that over 70% of hospitals would have thrombolysed.

4. Discussion

Our methods demonstrate how ML can be used alongside clinical registry data to identify where variation in decision making occurs. Specifically we have demonstrated that ML can be used to group hospitals according to their decision making and characterise patients that would have been treated differently had they attended a different hospital.

In our example we have used SSNAP, a clinical registry of in-hospital stroke care in the United Kingdom, to investigate variation in thrombolysis use for acute ischemic stroke in patients having a stroke out of hospital. By training separate RFs on patients from each of 132 hospitals, we found the RFs performed well on their own patients (mean accuracy 84.5%, mean AUC 0.90). Using these RFs, we assessed the amount of agreement between hospitals by putting all patients through all models, and we found that hospitals were much more likely to agree on patients not to thrombolysed than those to thrombolysed. Defining contentious patients as those in which there is less agreement between hospitals in

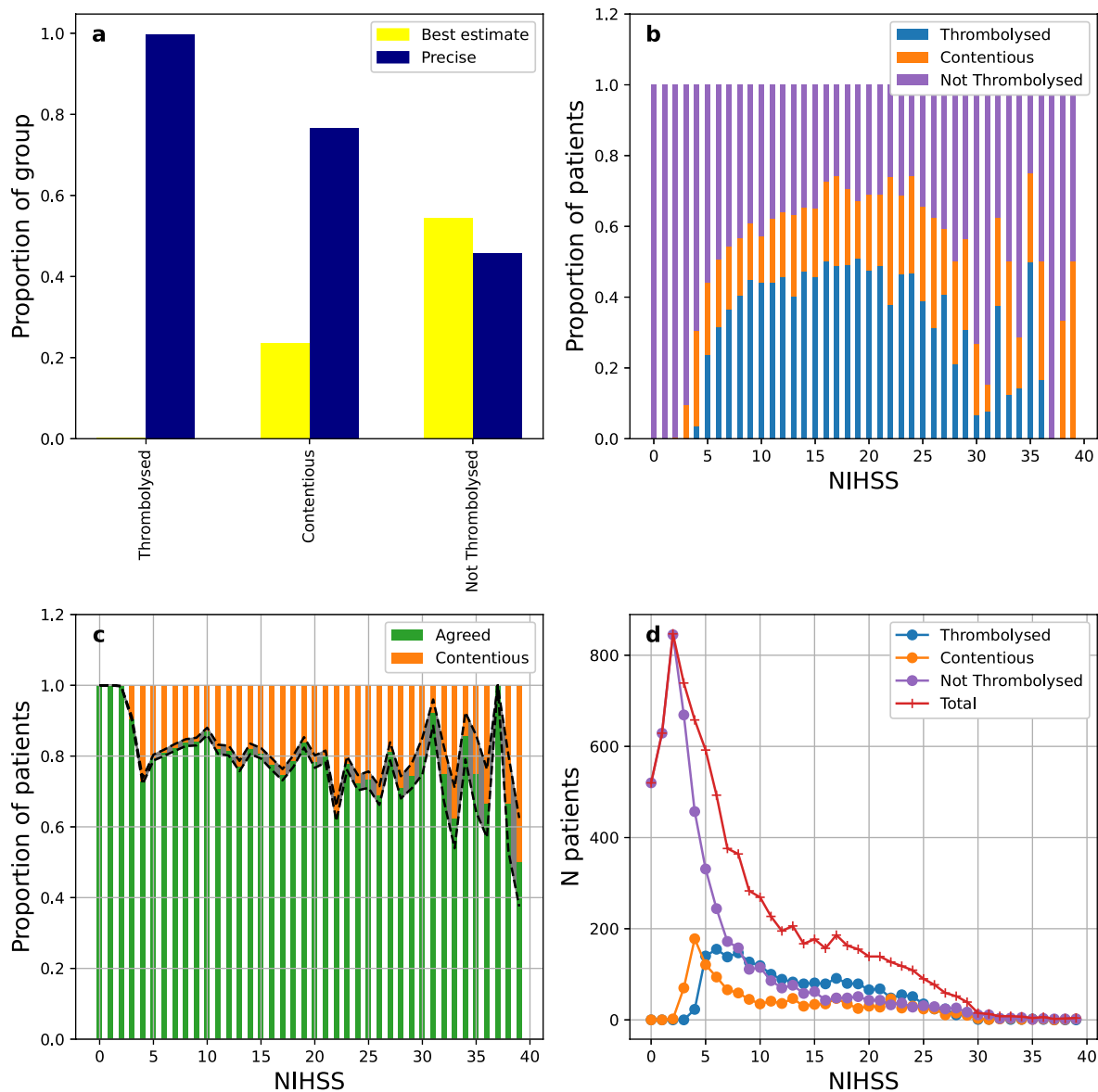


Fig. 5. **a** Distribution of onset time types for patients in a cohort of 10,000 that over 70% of hospitals would have thrombolysed (Thrombolysed), patients that over 70% of hospitals wouldn't have thrombolysed (Not Thrombolysed) and contentious patients. **b** Distribution of NIHSS values for patients in a cohort of 10,000 that over 70% of hospitals would have thrombolysed (Thrombolysed, blue), over 70% of hospitals wouldn't have thrombolysed (Not Thrombolysed, purple) and contentious patients (orange). **c** Distribution of NIHSS values for patients that over 70% of hospitals agreed on how to treat (Agreed, green) and contentious patients (orange). **d** Distribution of NIHSS values for patients in a cohort of 10,000 (Total, red) and stratified by whether over 70% of hospitals would have thrombolysed (Thrombolysed, blue), over 70% of hospitals wouldn't have thrombolysed (Not Thrombolysed, purple) and contentious patients (orange). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

how to treat (30–70% of hospitals would have thrombolysed), we trained an additional RF to discriminate between contentious patients and patients that over 70% of hospitals would have thrombolysed. We found that contentious patients are principally characterized by uncertainty in time of stroke onset (and therefore time since stroke), milder strokes (as measured by the NIHSS) and higher levels of prior disability (as measured by the mRS). It is important to note that, though our model has high accuracy, this type of model should not be considered a means to evaluate individual thrombolysis decisions in stroke. Rather, the value of the model is in identifying general patterns of how thrombolysis use varies between hospitals.

In the United Kingdom the current rate of thrombolysis use is 11.1% of all stroke, which is lower than the target in the NHS Long Term Plan of 20% and there is significant variation between hospitals (0–24.5%). This proportion is also significantly lower than rates reported from large

registries in other European countries [21], which suggests that there are significant numbers of stroke patients in the UK who stand to benefit from thrombolysis but who are not being treated. The median proportion of patients in whom the time of stroke onset is precisely known is 31.9%, and in whom it is a 'best estimate' is 36.8%, with 31.4% classified with an unknown time of onset (SSNAP Annual Results Portfolio, 2020–21 www.strokeaudit.org). However, there is a remarkable variation between hospitals in these proportions, ranging from 0.4 to 73.3% in our three-year national sample, a degree of variation that is difficult to understand. The willingness of individual clinicians to estimate the time of onset may relate to their own perception of the balance of risk and benefit from thrombolysis, particularly among patients at either end of the scale of neurological severity and in those with a greater degree of pre-stroke disability. A previous study [15] investigated factors that influence variation in thrombolysis decision-making using a discrete

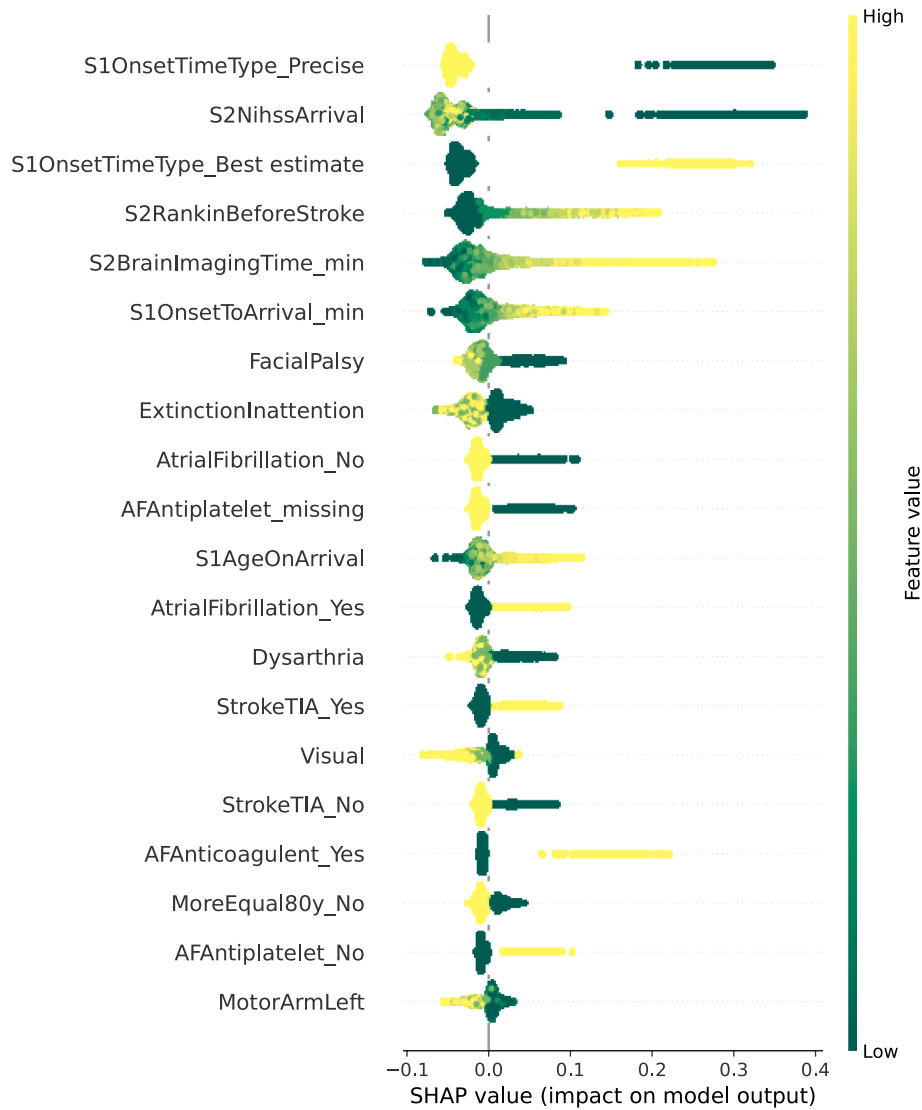


Fig. 6. SHAP summary plot for a random forest trained to classify whether a patient would have been thrombolysed at over 70% of hospitals (class 0) or is contentious (class 1).

choice experiment involving 138 clinicians, who were presented with hypothetical vignettes and asked how they would treat the patients described. This study also found that stroke severity and pre-stroke disability influenced a clinician's decision of whether or not to thrombolysed a patient. More recent imaging methods for acute stroke, using CT perfusion or MRI techniques, now offer the prospect of identifying patients eligible for thrombolysis according to a 'tissue window' as opposed to the more widely used 'time window' [22]. These developments would permit a more objective assessment of the 'tissue at risk' as the justification for thrombolysis at any time point after onset (known or unknown), and may enable clinical decision-making to become less variably based on individual clinicians' judgements about the accuracy of onset times or pre-stroke disabilities. Moving towards greater objectivity in deciding eligibility for thrombolysis may reduce the significant and unexplained variation in practice that we have highlighted in this study, which persists over more than 25 years since the first publication of the evidence for thrombolysis in acute ischemic stroke [23]. Identifying and characterizing the types of patients in whom eligibility is contentious also suggests areas where further clinical research is appropriate, and may help to improve the training of clinical decision-makers in thrombolysis, and thereby increase the consistency in thrombolysis use in the future.

4.1. Study limitations

When investigating the calibration of the hospital models we found that hospitals with both lower thrombolysis rates and fewer patients than average had less well calibrated models. This highlights the fact that when using an ML approach its important to consider whether there is both a sufficient amount of data in total (number of patients) and sufficient data in each class (thrombolysis rate) for the algorithms to perform well.

Performance of ML algorithms is also dependent on the level of detail contained in the data being used. We have demonstrated that hospital models perform well on their own patients, however it is likely that there are parameters that are not measured in SSNAP which are also significant contributors to variation in thrombolysis practice. It is therefore possible that parameters not present in the data, such as organizational factors like the availability and proximity of a CT scanner, or the availability of a stroke physician, impact model performance.

Our results show that when a stroke is severe (high NIHSS) the proportion of patients that are contentious is greater and shows more variability than for mild strokes (Fig. 5c). While the variability can be explained by only having a small number of patients with severe stroke in the data, it may also be a result of clinical decision-making.

Thrombolysis is more effective and has clearer outcomes when a stroke is mild to moderate, therefore the decision of whether to use thrombolysis for more severe strokes is likely to lead to variation: clinicians must make a judgement about whether the benefit of thrombolysis is worth the risk of hemorrhage. As severe strokes are less common, it is not possible to investigate clinical decision making in detail for patients with high NIHSS using ML and our data. It is therefore important to note that our results are not necessarily representative of clinical decision making in the most severe cases.

Our predictions about differences in clinical decision-making are necessarily at site level. We pick up on general differences in attitude to thrombolysis between sites, but we cannot detect differences that exist between individual clinicians (as decision-making is the end-result of a process, and may be collective, it may be that decisions can never be fully assigned to an individual).

In the example presented we used random forest algorithms due to their robustness, and previous work [24] has shown that, for this specific example, RF performs comparably to a neural network, and is superior to a logistic regression. However, the methods we have presented are more general and can be used alongside any ML algorithm.

5. Conclusion

The use of machine learning as applied to large-scale national stroke registry data can improve our understanding of the sources of variation in decision-making between hospitals in critical areas of clinical practice such as thrombolysis in acute ischemic stroke. The learning derived from advanced analysis could be used to reduce clinical variation through more effective audit and feedback to participating sites, and increase both the consistency of decision-making and the overall use and population benefit from treatments. The wider adoption of machine learning techniques in other comprehensive clinical registries similarly offers the prospect of reducing clinical variation in many other important disease areas.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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